

cloud platforms while another 22 per cent will run on hybrid cloud platforms [<https://erpinnews.com/enterprise-workloads-will-be-in-the-cloud>].

Cloud computing must satisfy five essential principles — on demand services, access to a network, resource pooling, elasticity and measured services. To achieve these essential features, cloud computing provides three kinds of service models — Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS) and Infrastructure-as-a-Service.

What is driving the adoption of cloud services?

- Reduced capex and opex to deliver business services.
- It minimises IT costs by reducing delivery times and improves the quality of the app development process.
- Changing business model – enterprises will soon become integrators of the best-of-the-breed services through collaboration.
- New regulatory requirements driven by a global collaborative economy and a need to address open markets.
- A digital explosion that is raising the bar to deliver a better customer experience.
- Transformation and optimisation across different process stacks — sales, front-office, middle office and back office.

Characteristics of the open source cloud

The open source cloud has the following characteristics:

- No vendor lock-in and there is seamless integration of the enterprise applications, products and systems developed/deployed by different organisations and vendors.
- The source code will be made available for the community, for adopters and end users to study and modify the software and to redistribute copies of either the original or the modified version. Source code will also be free from any royalty.
- With no vendor monopoly, the use of free and open standards is possible.

With data transferability and open data formats, there are greater opportunities to share data across interoperable platforms.

- Adoption of open source software enhances the interoperability with other enterprise solutions because the reuse of recommended software stacks, libraries and components is possible.

Figure 1 depicts the characteristics of the open source cloud.



Figure 1: Open source cloud characteristics

Cloud computing layers using open source

Figure 2 depicts cloud computing layers and the respective open source products, tools and software that map to each layer of the cloud computing infrastructure.

The layers cover the cloud market, the cloud broker platform, cloud management, SaaS, PaaS, IaaS, the cloud platform, virtualisation software/management, and hardware, which are used across enterprise applications.

The following sections briefly describe the technologies to be adopted on the cloud, and the equivalent open source components and products.

The cloud platform: Cloud based open source technology enables enterprises to accelerate innovation, creating and deploying applications with speed and consistency.

Application platforms handle complex connectivity issues behind the scenes (including automatic documentation), letting users concentrate on creating the right workflows and business logic, but also making it easier and more cost effective for organisations to manage frequent change. It is imperative to ensure that the integration solution is optimised for connecting the new applications in the cloud environment with older applications, systems, services and databases, no matter where they are located.

Open Application Programming Interfaces (APIs): The application's architecture uses open APIs to enable quick and transparent integration with other enterprise applications and systems implemented by various organisations, thereby providing access to data and services and promoting user/



Figure 2: Cloud computing layers and open source